Decision Trees

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Decision Trees

Learning Objectives

In this module, we will learn about decision trees:

- Regression Trees
- Splitting Criteria
- Stopping Criteria
- Algorithms – CART, ID3, C4.5
- Advantages and Disadvantages
- Overfitting in Decision Trees

Splitting Criterion for Regression Trees

- Consider,

$X = \{x_1, x_2 \dots x_N\}$ $\boxed{N=10}$

$y_1 \quad y_2 \quad \dots \quad y_N$

X_m m_{th} node $= \{x_2, x_4, x_6\}$

$y_2 \quad y_4 \quad y_6$

X: Set of training samples

X_m: Set of all $\mathbf{x} \in X$ at node m (where the further split has to happen)

y^i: Ground truth for the i^{th} sample

- We define the following Boolean,

$$b_m(\mathbf{x}^i) = \begin{cases} 1 & if\ \mathbf{x}^i \in X_m \\ 0 & otherwise \end{cases}$$

- Let g_m be the estimated ~~output value of~~ node m, given by the mean of the ground truths (~~outputs) of~~ samples present at node m,

$$g_m = \frac{\sum_{i=1}^{10} b_m(x_i)\, y_i}{\sum_{i=1}^{10} b_m(x_i)} = \frac{y_2 + y_4 + y_6}{3}$$

$$g_m = \frac{\sum_i b_m(\mathbf{x}^i) y^i}{\sum_i b_m(\mathbf{x}^i)}$$

In the case of noisy data, the median could work better than the mean.

Note: The estimated (predicted) value for all the samples belonging to a node must be the same. Take intuition from Classification Trees – All samples on the leaf node belong to the same class.

Splitting Criterion for Regression Trees

- In regression, the goodness of the split is measured by the mean squared error (MSE) from the estimated value.

$$MSE_m = \frac{1}{N_m}\sum_i \left(y^i - g_m\right)^2 b_m(x^i)$$

$$MSE_m = \frac{1}{3}\left\{(y_2-g_m)^2 + (y_4-g_m)^2 + (y_6-g_m)^2\right\}$$

$$g_m = \frac{1}{3}\left[y_2 + y_4 + y_6\right]$$

where,

$$N_m = |X_m| = \sum_i b_m(x^i)$$

- When using the mean as the GT, the MSE corresponds to the variance at node m.
- If MSE_m resides within the limits of the acceptable error, i.e., $MSE_m < \theta_r$, then m becomes a leaf node with g_m as its prediction.
- This creates a piecewise constant approximation with discontinuities at leaf boundaries.

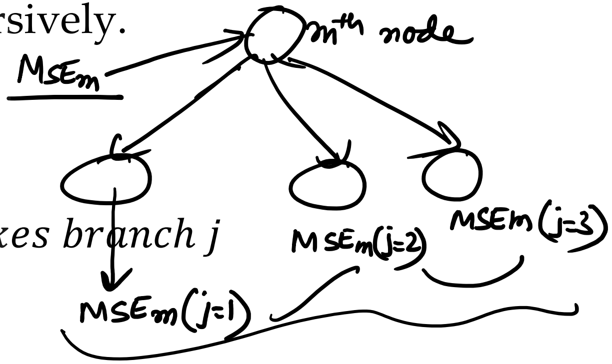
$$\sigma^2(\boldsymbol{y}) = \frac{\Sigma_{\mathrm{i}}\left(y_i - \mu_y\right)^2}{N}$$

Splitting Criterion for Regression Trees

- If the error is not acceptable, data reaching node m is split further such that the sum of the errors in the branches is minimum.
- Similar to the classification case, look for the attribute (or split threshold in the case of a numeric attribute) that minimizes the error and continue recursively.
- Let

$$X_{mj}: \text{Subset of } X_m \text{ taking branch j}$$

- We define the following,



$$b_{mj}(\boldsymbol{x}) = \begin{cases} 1 & if\ \boldsymbol{x} \in X_{mj}: \boldsymbol{x} \text{ reaches node } m\ and\ takes\ branch\ j \\ 0 & otherwise \end{cases}$$

$$g_m = \frac{\sum_t b_{mj}(\boldsymbol{x}^t) y^t}{\sum_t b_{mj}(\boldsymbol{x}^t)}$$

$$MSE_{m\_s} = \frac{1}{N_m} \sum_{j=1}^{3} \sum_t (y^t - g_m)^2 b_{mj}(x^t)$$

- The drop in the error for any split is given by the difference between MSE_m & MSE_{m_s}.

Stopping Criteria

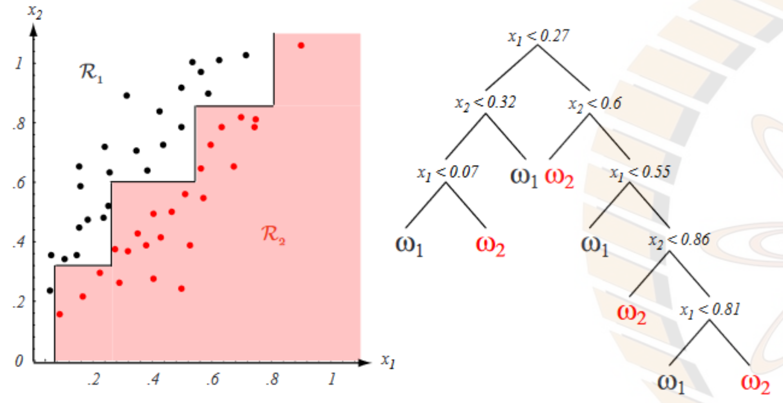
- If we continue expanding the decision tree until each leaf node represents the lowest possible impurity, it often indicates **overfitting** of the data, where each leaf node captures only a single pattern.
- On the other hand, if we stop splitting too early, the error on the training data remains high, leading to poor performance.
- To address these issues, use:
  - **Validation set**: Continue splitting until the error on the validation set is minimum
  - **Cross-validation**: Use multiple subsets of the data independently
- We can set a threshold for stopping the splitting process based on different factors, e.g., when a node has a fixed percentage of the total training set, stop splitting it further.

Decision Tree Algorithms

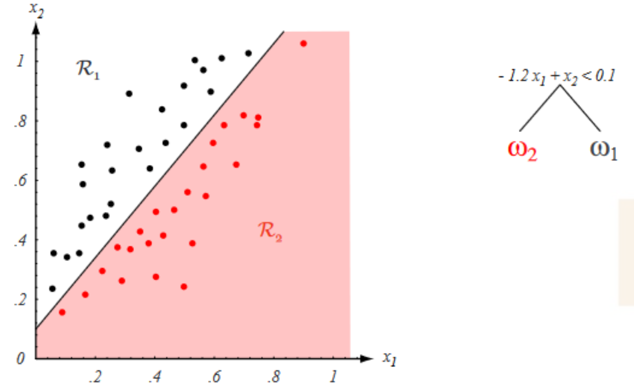
	CART	ID3	C4.5 J48
	Classification and Regression Trees	Iterative Dichotomiser 3	Extension of ID3
Year	Late 1980s	Late 1970s	Early 1990s
Task	Regression and Classification	Classification	Classification
Type of input data	Continuous & nominal	Categorical	Continuous & categorical
Speed	Average	Low	Faster than ID3
Pruning	Post-pruning	Not inbuilt	Pre-Pruning

Importance of Choice of Features

(a)



(b)



- If the class of node decisions does not match the form of the training data, the resulting decision tree will be complicated (see (a)).
- Here, decisions are parallel to the axes while, in fact, the data is better split by boundaries along another direction.
- If "proper" decision forms are used (here, linear combinations of the features), the tree can be quite simple (see (b)).

Hart, Peter E., David G. Stork, and Richard O. Duda. *Pattern classification*. Hoboken: Wiley, 2000.

Pruning

- Often, stopping tree splitting suffers from insufficient looking ahead
- A stopping condition may be met too early for overall optimal recognition accuracy
- Pruning is the inverse of splitting
- It could be of two types:

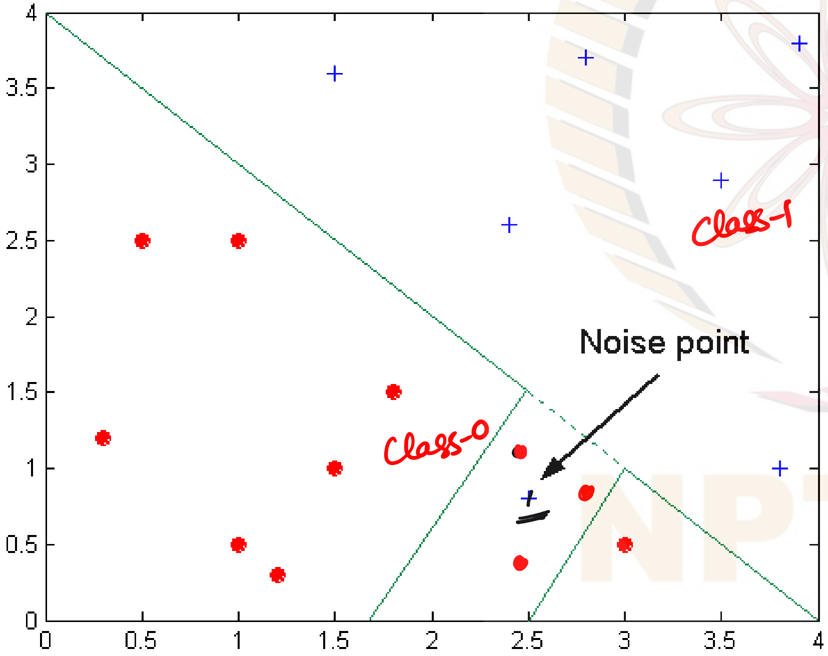
1. **Pre-pruning** [or Early stopping]: Tries to prune branches when the tree is being grown. E.g., when a validation set is available, a branch will not be grown if the validation error will increase by growing the branch
2. **Post-pruning**: Re-examines fully grown trees to decide which branches should be removed [More common, often simply referred to as *pruning*]

Post-Pruning

- **Procedure:**
  1. Allow the tree to grow fully—until leaf nodes have minimum impurity.
  2. Consider all pairs of leaf nodes (with a common antecedent node) for elimination
  3. Any pair whose elimination yields a small increase in impurity, can be eliminated and the common antecedent node is declared as a leaf node.

Overfitting due to Noise

- Decision boundary is distorted by noise.



Let us Try!

Consider the following $n = 16$ points in two dimensions for training a binary CART tree using the entropy impurity. Draw two trees (a) the training sample marked as '*' is used (b) the training sample marked as '†' (slightly shifted sample). Use CART algorithm.

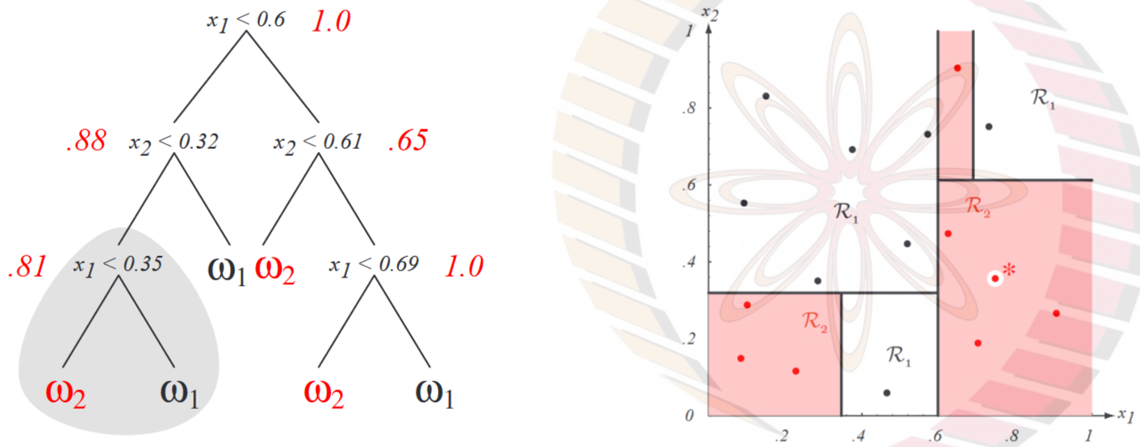
ω_1 (black)	
x_1	x_2
.15	.83
.09	.55
.29	.35
.38	.70
.52	.48
.57	.73
.73	.75
.47	.06

ω_2 (red)	
x_1	x_2
.10	.29
.08	.15
.23	.16
.70	.19
.62	.47
.91	.27
.65	.90
.75	$.36^*$ ($.32^\dagger$)

Hart, Peter E., David G. Stork, and Richard O. Duda. *Pattern classification*. Hoboken: Wiley, 2000.

Let us Try

(a)

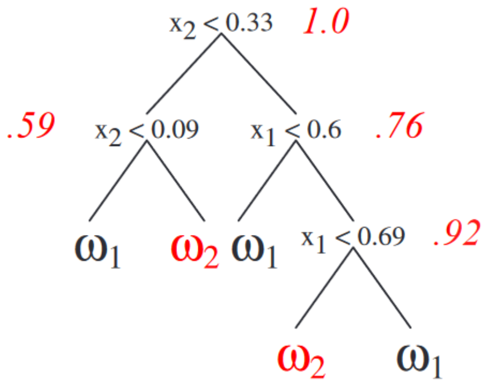


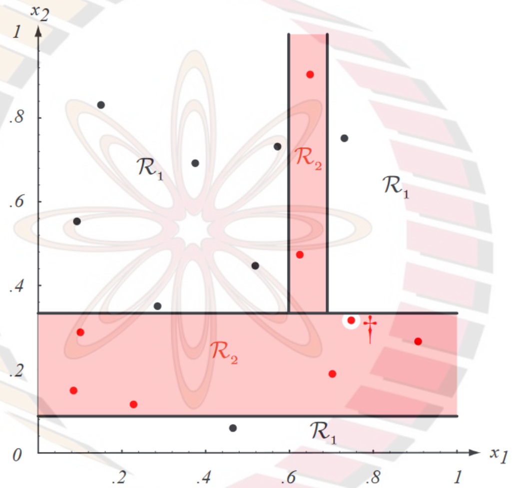
- Entropy impurity at nonterminal nodes is shown in red, with impurity at each leaf node being 0.

Hart, Peter E., David G. Stork, and Richard O. Duda. *Pattern classification*. Hoboken: Wiley, 2000.

Let us Try

(b)





- Entropy impurity at nonterminal nodes is shown in red, with impurity at each leaf node being 0.
- Notice the instability or **sensitivity of the tree** to training points: Alteration of a single point leads to a very different tree; due to CART's discrete & greedy nature.

Hart, Peter E., David G. Stork, and Richard O. Duda. *Pattern classification*. Hoboken: Wiley, 2000.

Advantages & Disadvantages

- **Advantages**:
  - Easy to understand and interpret: Expressed as a logical expression or IF-THEN conditions
  - Can handle both categorical and numerical data
  - Can be used for both binary and multi-class classification
  - Can handle large datasets with high accuracy
  - Can handle missing values
  - Can be used for feature selection, as they can identify the most important features for a given task
  - Can be used in ensemble methods ⇒ Random forest

- **Disadvantages**:
  - Prone to overfitting
  - Can be sensitive to small variations in the data
  - May not work well with imbalanced datasets

Decision Trees (Cont.)

- **Occam's Razor**: For a given training set, there exist many trees that code it without any error. For simplicity, we aim to find the smallest (low number of nodes) and least complex trees in a reasonable time.
- **Greedy search**: Learning a decision tree is a greedy algorithm since, at each step, we look for the best split at that step
- **Applications**: Used in many real-world applications, such as healthcare (e.g., predicting disease risk based on patient data), finance (e.g., predicting credit risk based on financial data), and marketing (e.g., predicting customer behavior based on demographic data), making them valuable skills for job opportunities

Overfitting in Decision Trees

To avoid overfitting

1. Pruning
2. Using the Validation set
3. Applying Cross-Validation
4. Ensemble of Decision Trees ➔ Random Forest

To Summarize

In this module, we learned about:

- Regression Trees
- Splitting Criteria
- Stopping Criteria
- Algorithms – CART, ID3, C4.5
- Advantages and Disadvantages
- Overfitting in Decision Trees

Reference: Introduction to machine learning / Ethem Alpaydin